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Semiconductors | North America

A defense of US memory stocks

We see the recent selloff as a healthy pricing in of durability concerns - capex, demand destruction, productivity etc. - and yet we see the strength as more durable than the market thinks, with memory supply remaining a gating factor for AI. Reiterate OW on MU, SNDK.

Key Takeaways

- In a normal cycle, flat spot price, capex increases, productivity might point to cycle peak - and those things have happened and have lead to profit taking
- But this is anything but normal - memory is a bottleneck, increasingly THE bottleneck, to AI builds (and agentic CPU builds) which appears durable
- Result - shortages are intensifying & customers are prepaying for large volume deals given conviction that these shortages will be sustainable.
- Second derivative deceleration is inevitable but at 4x earnings duration is all that matters and those indicators all appear positive
- KV cache optimization - weighing on the stock today - is just normal course productivity improvement.

As we highlighted before Micron earnings last week, it has been clear that there is very positive sentiment on near term results, and very low conviction about the durability of that strength. The desire to take profits has been evident for a while, and is healthy given the multiple of book value, and high valuation vs anything we have seen historically. Following a historic playbook has lead to selling, as gross margins approach all time highs, demand is negatively impacted by shortages, and materially higher capex signals some acceleration in supply growth. We monitor those concerns, take them seriously, but see demand strong enough to overtake them. There are also concerns that we take less seriously - flattening spot prices at levels so far above contract as to be less meaningful, and second derivative concerns that seem less relevant when 1q prices are up close to 90%.

But our view is that looking for sell signals from prior cycles misses the point. Memory is not just constrained by AI demand - memory is increasingly THE primary constraint on AI demand. For three years, we heard others make that case and we disagreed with it, as there was clearly slack in the DRAM supply, but that slack is gone. AI is consuming so much DRAM that there isn't enough left over for other sectors, and everywhere we look we see indications that it is a true bottleneck. It's holding back PC builds and smartphone builds. Cloud customers are paying premiums to the expected 2q price for 6 week expedites; do we think those customers are paying those premia to stockpile memory in a warehouse?

Higher capex will drive more supply, to be sure, but we aren't talking about a PC smartphone and server market growing 3-5% as the primary demand driver. AI

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Industry View **Attractive**

spending growth north of 50% - perhaps well north - based on processor company commentary, matters more each year as AI is a bigger portion of the pie. HBM4 complexity will absorb capacity, with content doubling next year with Rubin Ultra - but there is also exceptional demand for low power DDR5 for racks, and explosive demand in enterprise storage.

So we don't think 81% gross margin is the new normal, but it's also hard to see that getting worse in the next several quarters - perhaps even through the next two years. Two years of earnings at an $80+ run rate certainly suggests higher through cycle earnings power even with the assumption of cyclicality thereafter - and the $150 bn or so of free cash generation that comes with that can be put to productive use. Chips act restrictions limit cash returns, so if another use of cash does not emerge, both SNDK and MU company will de-lever and amass a substantial cash balance. We have long argued that matching cash flows with repurchases is suboptimal, as it means you are always buying high, so these restrictions might be positive.

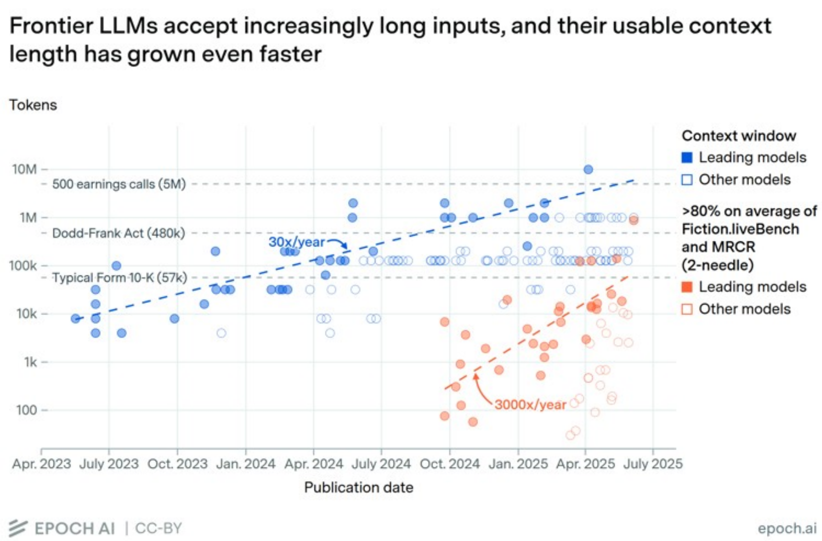
What about Google's "TurboQuant" memory optimization? In April of 2025 (here), Google wrote a white paper about "Turboquant", essentially a data compression algorithm which allows more data storage in the "key value cache" for frequently used data, with no impact to model performance. The company followed up that announcement with a blog post today saying that the company will be introducing the technology soon. It is being widely reported that Google has reduced memory usage by 6x, which leaves out that they are just talking about KV Cache memory, not memory overall. Memory stocks sold off again, at least partly due to this hyperbole.

Our take, after talking to industry folks on this today, is that this is an evolutionary development, with basically no surprises for memory. Memory usage is one of the key determinants of AI performance, which makes memory a key aspect of evolutionary improvements. Of course Google is one of the most innovative companies in this space, so an incremental improvement is not particularly surprising, and compression algorithms for KV cache have been a focus for years, per our contacts - "and the biggest improvements are not advertised" since they can be an edge in the business. It's also worth noting that Google's Gemini 3 and 2.5 Pro models feature a 1M token context window, but the company has disclosed that they tested up to 10M tokens with great results with Gemini 1.5 pro but choose not to release the model in part due to inference costs (here). So we would expect that as costs come down due to innovations like these and others, it will be used to serve higher intelligence more compute intensive products.

Further, more efficient use of KV Cache doesn't have much direct impact. Most frequently the KV cache is held in high bandwidth memory, the content of which is fixed and cannot change; if there is additional caching offload required, it is typically moved to the "unified address space" - ie the 18 TB of LPDDR5 memory in the rack, which is also fixed. If there is need for tertiary offload, It can be in an SSD or HDD, and NVIDIA has talked about a tier of SSD KV cache storage which we think starts next year. More efficiency could have impact on that 3rd tier, but generally commentary is that the improvements are mostly designed to allow for extended context windows - higher capability - rather than trying to reduce tertiary storage costs. There's just no indication that demand for memory or storage is going down, universally across our contacts.

NVIDIA has described that they can deliver 1 million x performance improvement per decade, and that is not unrealistic - and is partly improvement from the ecosystem, which in this market drives a constant stream of innovations. Occasionally those disruptions create some angst that chips are becoming too efficient, but it's central to the growth of the ecosystem.

Exhibit 1: Even at face value a 6x reduction in KV cache memory usage becomes much less meaningful in the context of 30x annual growth in context windows



Source: EpochAI

Should we be worried about Open AI's shutdown of Sora? Open AI reportedly (here) has decided to suspend its SORA AI video generation app, which lead to significant hand wringing. Our view is that this is clear evidence of our investment themes, that there simply is not enough compute for double digit weekly growth in tokens. There is too much demand, not enough supply. Profit concerns may also play a role, but the primary issue is the compute shortage. That said, video is also very storage intensive, so the shortage of memory and HDD storage may also be a factor. But we have a hard time being overly negative for commodity products that demand is too good to be fully served, especially at these valuations.

What about long term contracts? After Micron reported, we had said that we had hoped to get more color on structure of long term deals - but that is coming, as we are getting reports that some of these deals will involve multiple billions in prepayments. Even if Micron is reluctant to describe individual deals, we will see cash & revenue reserves on the balance sheet. We think that's important - less so that deals will be ironclad, as everything is negotiable, but more the signalling that customers are convinced that the memory shortage is a multiple year concern.

One additional thing to consider - what's the best way to play the recovery in general purpose servers? Since September we have been writing about a surge in general purpose servers, with agentic AI driving growth in CPU usage. Microprocessors are in short supply, with rising prices, and both AMD and Intel have been strong on this theme.

Its our view that the best way to play growth in general purpose servers will be memory. AMD and Intel should see some benefit from rising prices, but the increases are measured - processors are not a commodity. Both stocks have higher multiples pricing in prospects for Intel in Foundry, and AMD in AI, which to us remain less certain; buying memory stocks at these levels seems a higher leverage way to play this agentic strength.

The wall of worry clearly in evidence should provide a good setup, with stocks at mid single digit multiples on numbers that are not sustainable forever, but are sustainable for several quarters. Annual cash generation at these earnings levels for both Micron and Sandisk can be 15-25% of current market cap - the exact amount dependent upon scale of prepayments - and while it won't last forever, it is going to last for long enough to see the stocks move materially higher, in our view.

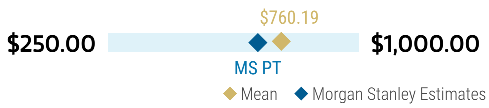
Risk Reward – SanDisk Corporation. (SNDK.O)

NAND improving quickly on the back of accelerating cloud demand

PRICE TARGET $690.00

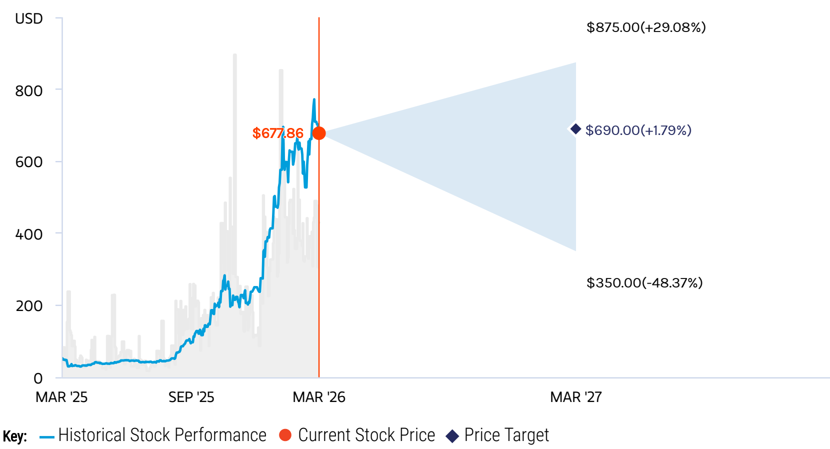
We assume 23x through-cycle EPS of $30.00 (vs. our estimate of the trailing 9-year average of $6.33). As improving industry dynamics as well as eSSD demand lead to higher levels of profitability than we have seen over the last few years, 23x is a discount to our through-cycle multiple target for MU (25x) as the lower AI exposure is offset by historically higher FCF conversion

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART

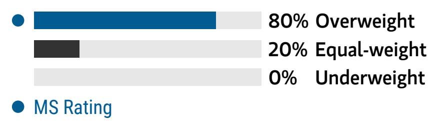


Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

AI demand is in the process of transforming the NAND market, and driving peak cycle EPS to multiples of what we have seen in the past. With still minimal new investment in capacity and evolving AI demand drivers we see a long runway for upward revisions.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE $875.00

25x through-cycle EPS of $35

Our bull case of $800 per share represents 25x a higher through-cycle EPS of $32.00. Here earnings sustain above to prior peak cycles ($20+ per share) and more like the current environment. As well as with more mild downturns, higher levels of profitability supporting a higher multiple.

BASE CASE $690.00

23x through-cycle EPS of $30.00

Our base case of $690 is based on 23x through-cycle EPS of $30.00 (vs our estimate of the trailing 9-year average of $6.33). As improving industry dynamics as well as eSSD demand lead to higher levels of profitability than we have seen over the last few years, 23x is a discount to our through-cycle multiple target for MU (25x) as the lack of direct AI is offset by historically higher FCF conversion.

BEAR CASE $350.00

~16x through-cycle EPS of $22.

We think Sandisk can avoid much of the pain of the last downturn ($13 per share loss), and with a better upcycle, with the multiple at the low end of the semis group where more capital-intensive cyclical companies tend to trade.

Risk Reward – SanDisk Corporation. (SNDK.O)

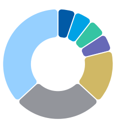
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
GAAP Revenue ($, mm)	7,355	15,499	24,989	24,456
Non-GAAP Gross Margin (%)	30.3	60.2	73.4	70.8
Non-GAAP EPS ($)	2.74	41.09	82.73	71.92
Inventory ($, mm)	2,079	1,991	2,182	2,335
DOI	145.5	115.9	117.7	117.5

INVESTMENT DRIVERS

- Improved pricing and demand strength drive earnings growth
- Position in DC SSD market improves

GLOBAL REVENUE EXPOSURE



- 0-10% Japan
- 0-10% Latin America
- 0-10% Mainland China
- 0-10% UK
- 10-20% Europe ex UK
- 20-30% North America
- 30-40% APAC, ex Japan, Mainland China and India

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

5/5 **BEST**	24 Month Horizon	5/5 **MOST**	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Higher NAND content growth from edge AI applications
- Quicker eSSD penetration in the datacenter
- Sandisk's investments in advanced memory technologies such as HBF (high Bandwidth Flash) pay dividends

RISKS TO DOWNSIDE

- NAND industry growth disappoints
- Capex growth returns as industry participants invest to gain share
- Sandisk loses market share as they fail to gain traction in datacenter
- China continues to gain share

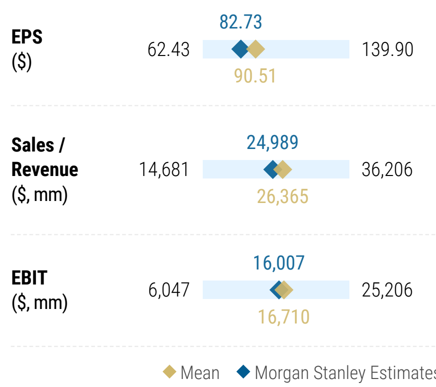
OWNERSHIP POSITIONING

Inst. Owners, % Active	53.3%
HF Sector Long/Short Ratio	2.1x
HF Sector Net Exposure	26.4%

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Jun 2027e



Source: Refinitiv, Morgan Stanley Research

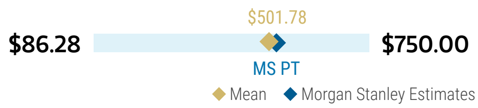
Risk Reward – Micron Technology Inc. (MU.O)

See multiple quarters of upward revisions, with AI driving a higher multiple

PRICE TARGET $520.00

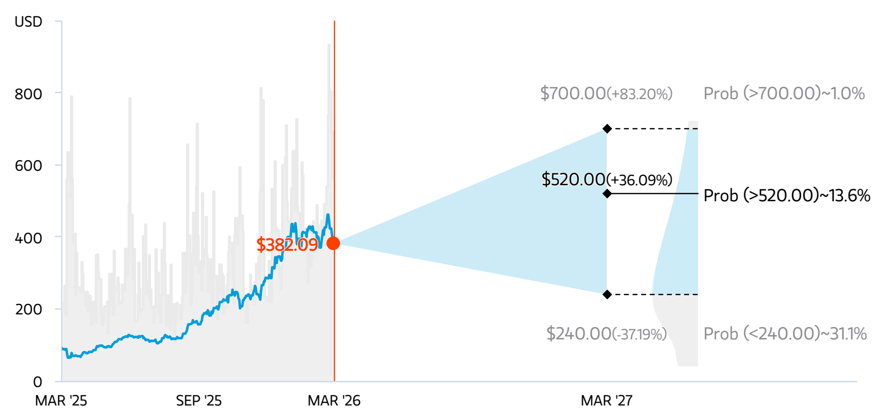
~25x through-cycle earnings of US $21.00, a premium to history reflecting new opportunities in AI, At the high end of broader semis.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



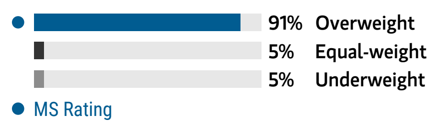
Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 25 Mar 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology here

OVERWEIGHT THESIS

- DRAM fundamentals are in uncharted territory, and should continue to improve as datacenter/AI continue their upward trajectory
- Execution on AI is underappreciated, and we expect Micron to maintain share in CY26 vs the competition. Supporting margins and driving a higher multiple than prior cycles
- Cycle longevity will be key, and S/D may stay tight for 2-3 more years

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: *Positive*
Secular Growth: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE $700.00

28x through-cycle earnings of US$25.00

Gross margin improvement continues, driven by scale, AI mix, and cost improvements in new products. Pricing pressure alleviates as demand sustainably moves above supply driven by HBM's wafer intensity, a product category where MU cements performance leadership on future products.

BASE CASE $520.00

~25x through-cycle earnings of US$21.00

Our through-cycle earnings estimate of US$21.00 is a premium to average earnings over the last 8 years due mostly to AI. Our 25x multiple reflects the market's enthusiasm for the AI opportunity, and at the midpoint of the semis group

BEAR CASE $240.00

16x through-cycle earnings of US$15.00

Memory begins to enter a downturn in early 2027. As the strength attributed to demand in the early part of the year ended up being inventory build at customers. Multiple compresses severely after an underwhelming peak

Risk Reward – Micron Technology Inc. (MU.O)

KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
GAAP Revenue ($, mm)	37,378	110,489	150,620	138,790
Non Gaap Gross Margin (%)	40.9	77.8	82.3	78.8
Non-GAAP EPS ($)	8.29	59.36	81.36	72.32
Inventory ($, mm)	8,355	8,564	9,100	10,164
DOI	133.7	123.5	120.6	122.4

INVESTMENT DRIVERS

- Improved pricing and demand strength drive earnings growth

GLOBAL REVENUE EXPOSURE

0-10%	APAC, ex Japan, Mainland China and India
0-10%	India
0-10%	Japan
0-10%	MEA
0-10%	Mainland China
0-10%	UK
10-20%	Europe ex UK
40-50%	North America

Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

5/5 **BEST**	24 Month Horizon	3/5 **MOST**	3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Customers continue to demonstrate an appetite to take on inventory around macroeconomic uncertainty
- Additional wafer intensity of HBM further improves overall supply and demand
- MIcron's HBM share surpasses expectations

RISKS TO DOWNSIDE

- Pricing can turn quickly; a falter in end demand with inventories elevated could lead to a swift price reduction
- HBM demand falters and competition intensifies, pressuring pricing

OWNERSHIP POSITIONING

Inst. Owners, % Active	50.1%
HF Sector Long/Short Ratio	2.1x
HF Sector Net Exposure	26.4%

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure – Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Oct 2026e

	Low	Morgan Stanley Estimates	Mean	High
Sales / Revenue ($, mm)	81,813	110,489	108,323	116,930
EBITDA ($, mm)	58,763	88,885	86,152	94,958
Net income ($, mm)	44,520	68,321	65,417	72,496
EPS ($)	37.54	59.36	57.19	63.91

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Valuation Methodology and Risks

Advanced Micro Devices (AMD.O)

Our $255 PT for AMD equals ~27x base case FY2027e MWEPS of $9.46, reflecting further share gains at the expense of Intel, and 43% y/y growth in datacenter (~80% in AI)

Risks to Upside

- Datacenter GPU outperforms expectations, closing the gap with Nvidia
- PC and server share gain accelerates; Intel's competitive response is less impressive than expected
- Server refresh drives datacenter revenue above expectations

Risks to Downside

- Intel's server CPUs in 2027 stifle AMD's momentum and allow it to regain share
- AMD loses graphics share to NVIDIA
- Datacenter GPU underperforms expectations

Intel Corporation (INTC.O)

~42x EPS CY2027 EPS of $0.97, 42x is above the high end of the large cap logic semi peer group, reflecting high leverage potential on numbers that are still depressed, and foundry optionality, despite our longer term skepticism

Risks to Upside

- Foundry partnerships de-risk the story and further improve the multiple
- The company regains lost share in desktop and server following CPU shortages

Risks to Downside

- AMD competition increasingly becomes more significant, which could lead to further share losses in processors and pressure on ASPs
- Minimal success in foundry leads to an inflated cost structure

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

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(as of February 28, 2026)

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For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1528	41%	444	49%	29%	690	42%
Equal-weight/Hold	1590	43%	374	41%	24%	725	44%
Not-Rated/Hold	4	0%	1	0%	25%	1	0%
Underweight/Sell	588	16%	87	10%	15%	220	13%
Total	3,710		906			1636	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

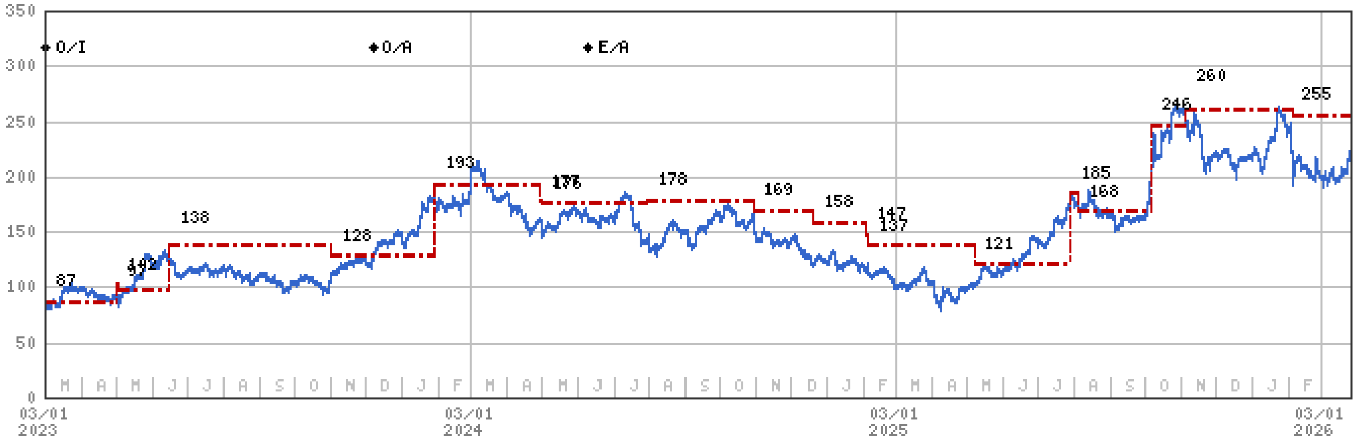
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Advanced Micro Devices (AMD.O) - As of 03/26/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 3/1/21 : NA/I; 3/10/21 : NA/I; 4/7/21 : NA/I; 6/22/22 : O/I; 12/7/23 : O/A; 6/9/24 : E/A

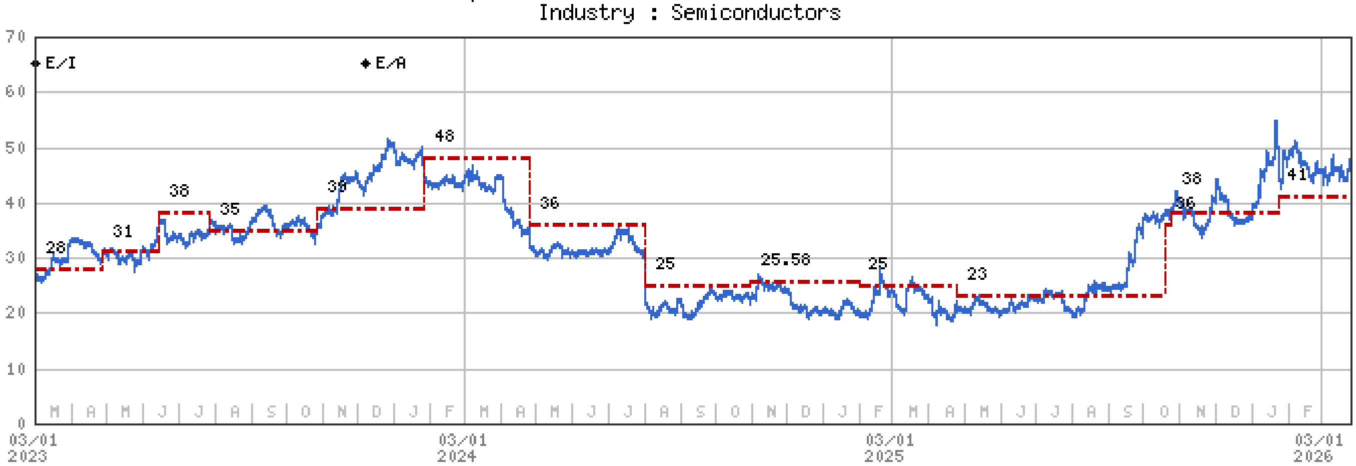
Price Target History: 10/27/20 : NA; 3/10/21 : NA; 4/7/21 : NA; 6/22/22 : 103; 8/1/22 : 101; 8/3/22 : 102; 9/23/22 : 95; 10/6/22 : 86; 10/31/22 : 77; 1/31/23 : 87; 5/1/23 : 102; 5/2/23 : 97; 6/15/23 : 138; 11/1/23 : 128; 1/29/24 : 193; 4/29/24 : 177; 4/30/24 : 176; 7/30/24 : 178; 10/29/24 : 169; 12/19/24 : 158; 2/3/25 : 147; 2/5/25 : 137; 5/7/25 : 121; 7/29/25 : 185; 8/5/25 : 168; 10/6/25 : 246; 11/4/25 : 260; 2/4/26 : 255

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Intel Corporation (INTC.O) - As of 03/26/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 3/1/21 : O/I; 10/22/21 : E/I; 3/3/22 : U/I; 2/22/23 : E/I; 12/7/23 : E/A

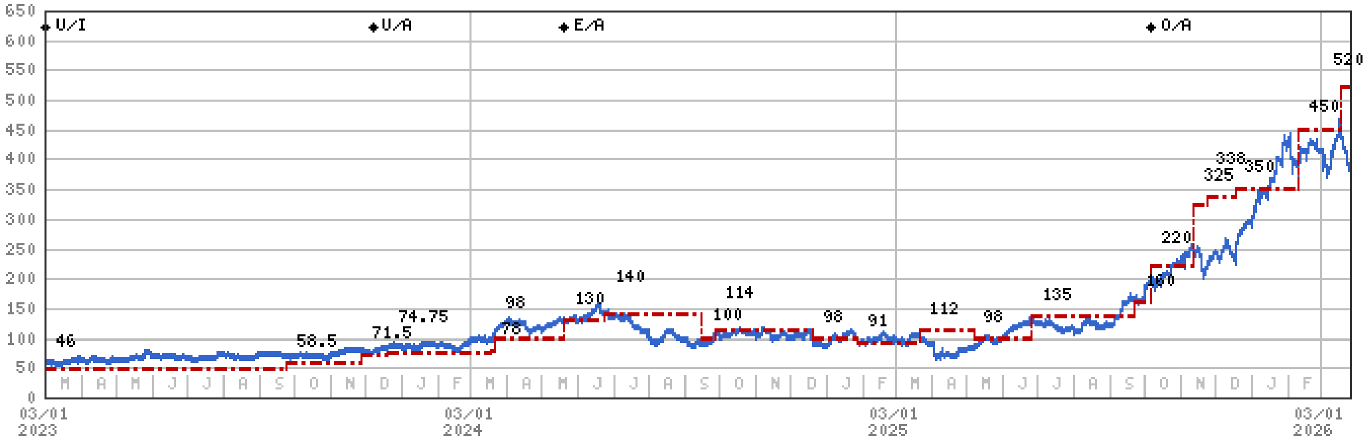
Price Target History: 1/22/21 : 72; 7/23/21 : 70; 10/5/21 : 67; 10/22/21 : 55; 3/3/22 : 47; 4/28/22 : 46; 7/28/22 : 36; 10/27/22 : 29.5; 2/22/23 : 28; 4/27/23 : 31; 6/15/23 : 38; 7/28/23 : 35; 10/27/23 : 39; 1/26/24 : 48; 4/26/24 : 36; 8/2/24 : 25; 10/31/24 : 25.58; 1/31/25 : 25; 4/25/25 : 23; 10/20/25 : 36; 10/24/25 : 38; 1/23/26 : 41

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Micron Technology Inc. (MU.O) - As of 03/26/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 3/1/21 : O/I; 8/12/21 : E/I; 7/21/22 : U/I; 12/7/23 : U/A; 5/20/24 : E/A; 10/6/25 : O/A

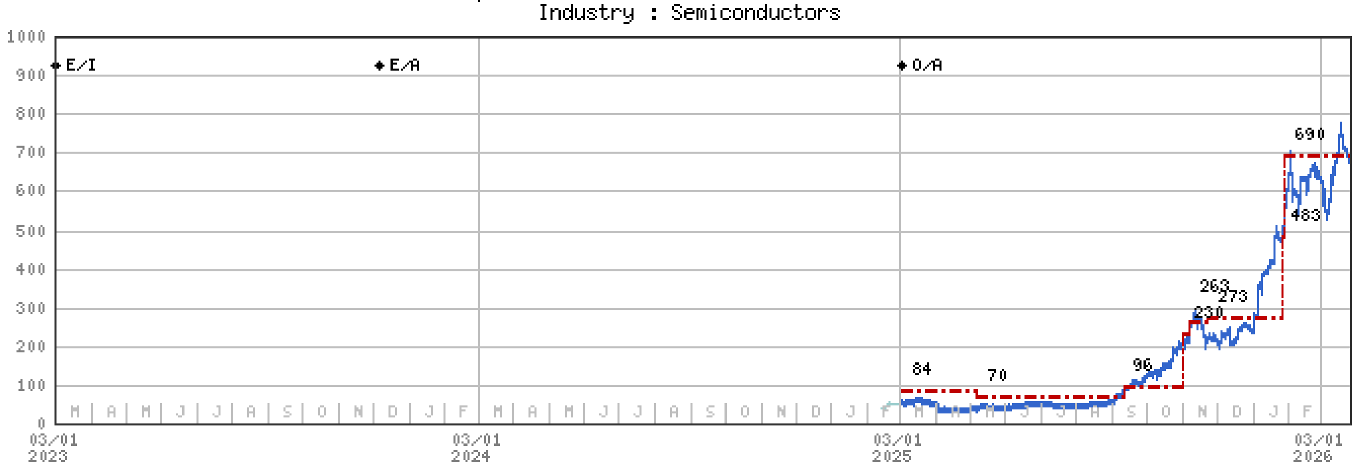
Price Target History: 1/4/21 : 88; 3/1/21 : 105; 8/12/21 : 75; 12/20/21 : 77; 3/29/22 : 83; 6/21/22 : 56; 9/30/22 : 49; 12/21/22 : 46; 9/24/23 : 58.5; 11/27/23 : 71.5; 12/20/23 : 74.75; 3/18/24 : 78; 3/21/24 : 98; 5/20/24 : 130; 6/23/24 : 140; 9/15/24 : 100; 9/26/24 : 114; 12/19/24 : 98; 1/27/25 : 91; 3/21/25 : 112; 5/7/25 : 98; 6/26/25 : 135; 9/22/25 : 160; 10/6/25 : 220; 11/12/25 : 325; 11/23/25 : 338; 12/17/25 : 350; 2/10/26 : 450; 3/18/26 : 520

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

SanDisk Corporation. (SNDK.O) - As of 03/26/26 GMT in USD
Industry : Semiconductors



Stock Rating History: 3/1/21 : E/I; 12/7/23 : E/A; 3/3/25 : O/A

Price Target History: 3/3/25 : 84; 5/7/25 : 70; 9/11/25 : 96; 11/2/25 : 230; 11/7/25 : 263; 11/23/25 : 273; 1/26/26 : 483; 1/29/26 : 690

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

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INDUSTRY COVERAGE: Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (03/25/2026)
Joseph Moore		
Advanced Micro Devices (AMD.O)	E (06/09/2024)	$220.27

Aeva Technologies Inc (AEVA.O)	E (07/19/2021)	$15.26
Allegro Microsystems Inc (ALGM.O)	O (02/13/2026)	$32.63
Ambarella Inc (AMBA.O)	O (03/29/2016)	$54.09
Amkor Technology Inc (AMKR.O)	E (11/08/2023)	$49.93
Analog Devices Inc. (ADI.O)	O (11/16/2023)	$322.03
Astera Labs Inc (ALAB.O)	O (05/11/2025)	$120.33
Broadcom Inc. (AVGO.O)	O (06/09/2024)	$318.81
GlobalFoundries Inc (GFS.O)	E (10/28/2024)	$46.74
Intel Corporation (INTC.O)	E (02/22/2023)	$47.18
IonQ Inc (IONQ.N)	E (04/25/2023)	$31.96
Marvell Technology Group Ltd (MRVL.O)	E (09/14/2015)	$98.45
Microchip Technology Inc. (MCHP.O)	E (07/10/2024)	$65.16
Micron Technology Inc. (MU.O)	O (10/06/2025)	$382.09
Navitas Semiconductor Corp (NVTS.O)	U (04/06/2025)	$9.48
NVIDIA Corp. (NVDA.O)	O (03/16/2023)	$178.68
NXP Semiconductor NV (NXPI.O)	O (02/11/2025)	$197.61
ON Semiconductor Corp. (ON.O)	E (05/11/2025)	$63.10
Qorvo Inc (QRVO.O)	E (10/28/2025)	$80.74
Qualcomm Inc. (QCOM.O)	U (02/10/2026)	$130.35
SanDisk Corporation. (SNDK.O)	O (03/03/2025)	$677.86
Semtech Corp. (SMTC.O)	E (04/06/2025)	$80.04
Silicon Laboratories Inc. (SLAB.O)	E (01/19/2021)	$207.07
Skyworks Solutions Inc (SWKS.O)	E (11/28/2018)	$56.19
Texas Instruments (TXN.O)	U (04/13/2020)	$196.77
Wolfspeed, INC (WOLF.N)	NR (04/06/2025)	$18.50
Lee Simpson		
Arm Holdings plc (ARM.O)	O (07/19/2024)	$157.07
Cadence Design Systems Inc (CDNS.O)	O (02/14/2024)	$281.39
Synopsys Inc. (SNPS.O)	E (02/27/2026)	$410.13

Stock Ratings are subject to change. Please see latest research for each company.

* Historical prices are not split adjusted.